VigiThink eTMF

Complete Industry Blueprint & Build Plan

React JS + Node JS

Reference systems: Veeva Vault eTMF | Medidata eTMF | ArisGlobal LifeSphere eTMF | Phlexglobal PhlexTMF | Montrium eTMF

Prepared for Clinical Research Training Program

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1. Regulatory & Industry Foundation

Every eTMF, regardless of vendor, exists to prove trial integrity, GCP compliance, and inspection-readiness. The system must be built around the following standards:

Field	What it does / why it matters
ICH E6(R2) / E6(R3) GCP	Defines the Essential Documents required before, during, and after a trial
DIA TMF Reference Model (v3.x)	Industry-standard taxonomy: Zones, Sections, Artifacts, Sub-artifacts
21 CFR Part 11 (US FDA)	Electronic records and signatures — audit trail, e-signature, access control
EU GDPR / EU Annex 11	Data privacy and computerized systems validation
ICH E8(R1) / ISO 14155	Quality by design; medical device trials
ALCOA+ / ALCOA-C	Attributable, Legible, Contemporaneous, Original, Accurate, Complete, Consistent, Enduring, Available
21 CFR Part 312 / 812	Sponsor and investigator recordkeeping obligations

Teaching point: an eTMF is a compliance and inspection-readiness engine, not a file store. Every field must answer — can an inspector use this system to prove the trial was run correctly?

2. Core Data Model — TMF Reference Model

All five reference vendors structure documents on the same hierarchy. The database must implement this exact structure:

- Zone (11 total, e.g. Zone 01 — Trial Management)
  - Section (e.g. 01.01 — Trial Oversight)
    - Artifact (e.g. Trial Management Plan)
      - Sub-Artifact / Document (the actual filed record with version history)

The 11 Standard TMF Zones

- Trial Management
- Central Trial Documents
- Central Trial Documents – IRB/IEC and Other Approvals
- IP and Trial Supplies
- Safety Reporting
- Site Management (country/site level)
- IRB or IEC and Other Approvals (site level)
- IP and Trial Supplies (site level)
- Safety Reporting (site level)
- Local/Central Lab, Imaging and Other Sample/Data
- Third Party / Vendor Oversight

Each Zone contains multiple Sections, each Section multiple Artifacts — 200+ artifacts total across the DIA v3.x model. The Artifact library should be seeded and fully configurable.

3. Complete Field List — Document & Record Level Metadata

This is the complete, field-by-field metadata schema. Every category below maps to a real regulatory or operational reason — nothing is decorative.

3.1 Identification Fields

Field	What it does / why it matters
document_id	Unique system-generated ID that every other table (QC, workflow, audit trail, e-signature) points back to — this is the primary key tying a document's whole life story together
tmf_zone_id / section_id / artifact_id / subartifact_id	Places the document at its exact address inside the DIA TMF Reference Model tree, so the tree navigator and completeness reports know exactly where it belongs
study_id / protocol_number	Scopes the document to one specific trial — almost every query and permission check in the system filters by this field first
country_code (ISO 3166)	Scopes the document to a country, which drives country-specific retention rules and regulatory authority requirements
site_id / site_number	Scopes the document to a specific investigator site within a country
sponsor_id / cro_id / vendor_id	Records which organization owns or is accountable for the document — needed when work is outsourced to a CRO or vendor
document_title	The human-readable name shown in lists and search results, independent of the internal file name
document_type / document_subtype	Classifies the record (Protocol, ICF, IB, CV, Certificate, Report, Correspondence) which in turn drives which QC checklist and workflow automatically apply

3.2 Version & Lifecycle Fields

Field	What it does / why it matters
version_number	Tracks major.minor version (e.g. 1.0, 2.0) so users always know if they are looking at the current or a historical copy
status	Draft / In Review / Approved / Effective / Superseded / Obsolete / Placeholder / Expected — this single field drives what actions are even allowed on the document (e.g. only Effective documents count toward completeness)
is_placeholder	True when the system knows a document should exist (from the Expected Document List) but the real file has not been filed yet — shows as a visible gap in the completeness report instead of silently missing
is_certified_copy	Distinguishes a true original from a certified copy, which matters for ALCOA 'Original' data integrity during an inspection
source_type	Original / Certified Copy / Scanned Copy — used together with is_certified_copy to prove chain of custody for paper-to-digital conversions
supersedes_document_id / superseded_by_document_id	Chains versions together so the system can always show 'this replaced that' and prevent an old version being mistaken for current

3.3 Date Fields

Field	What it does / why it matters
document_date	The date printed on the document itself — independent of when it was uploaded, and used to check the document was actually valid on the date it covers
effective_date / expiration_date / review_due_date	Defines the validity window of a document — e.g. an expired IB should trigger a renewal task automatically
upload_date_time	System timestamp of when the file entered the system, used purely for technical audit purposes
filing_date	The date the document was actually filed into the TMF — compared against document_date to calculate the filing-lag KPI that inspectors specifically look for
qc_review_date / approval_date	Marks when each workflow gate was passed, used to measure how long documents sit in review
archival_date / retention_end_date	retention_end_date is auto-calculated from the country's regulatory retention rule and archival_date; together they tell the system exactly when a record becomes eligible for destruction (unless a legal hold is active)

3.4 People & Role Fields

Field	What it does / why it matters
author_id / uploaded_by_id	Records who wrote the document versus who physically uploaded it into the system — these are often different people and both matter for accountability
qc_reviewer_id / approver_id	Identifies exactly who signed off at each workflow stage, which is what an inspector will ask to see
document_owner_id / custodian_id	The person accountable for the document's accuracy over its lifetime, separate from whoever happens to be editing it today
last_modified_by_id	Always shows who touched the record most recently, useful for quickly finding who to ask about a recent change
signatory_id(s)	Stores every person who electronically signed the document, supporting multiple signers on one record

3.5 Quality Control (QC) Fields

Field	What it does / why it matters
qc_status	Not Started / In Progress / Passed / Failed / Query Raised — this is the field the QC queue screen filters and sorts by
qc_checklist_id	Points to the specific checklist (completeness, legibility, correct version, correct signatures) that was actually used, so QC decisions are reproducible and defensible
qc_comments / quality_issue_flag / quality_issue_type	Captures exactly what was wrong (missing signature, wrong version, illegible scan, misfiled) so the same mistake can be tracked and trended over time

Field	What it does / why it matters
qc_score / health_indicator	Rolls QC results up into the Red/Amber/Green status that powers the dashboard heatmap, so a manager can see study health without opening a single document

3.6 Compliance & Access Fields

Field	What it does / why it matters
confidentiality_level	Public / Internal / Restricted / Highly Confidential — this is the field the access control engine reads before deciding whether to even show the document in search results to a given user
access_control_list	The actual list of roles/users allowed to view this specific record, used when a document needs tighter restriction than its default role permissions
redaction_status	Tracks whether personal data (PII/PHI) has been removed from a document before it can be shared externally or with a broader audience
is_essential_document	Flags the document as one of ICH E6's required Essential Documents — these are the ones completeness reporting and inspection-readiness scoring actually count
regulatory_binding_flag	Marks a document as a hard blocker for a milestone — e.g. the system will not let 'Site Activation' be marked complete until every regulatory_binding_flag document for that site is Effective
signature_type	Wet ink (scanned) / Electronic (Part 11) / Third-party e-sign — determines what proof the system must store alongside the document to make the signature legally defensible

3.7 File & Technical Fields

Field	What it does / why it matters
file_name / file_format / file_size	Basic file attributes used for display, download, and storage management
checksum_hash (SHA-256)	A fingerprint of the file's exact bytes — if the hash ever changes without a new version being created, the system knows the file was tampered with
storage_path / object_storage_key	The actual location of the file in S3-compatible storage, kept separate from the metadata database so files and data can scale independently
ocr_text	Machine-extracted text from scanned documents, which is what powers full-text search across image-based PDFs
language_code / page_count / thumbnail_path	Supports multi-language filtering, quick page-count checks during QC, and fast-loading thumbnails in document grids

3.8 Relationship & Traceability Fields

Field	What it does / why it matters
related_document_ids[]	Links documents that belong together conceptually (e.g. a cover letter and its approval), so opening one surfaces the other

Field	What it does / why it matters
milestone_id	Connects a document to the trial milestone it supports, which is what lets the milestone tracker show real-time readiness
related_task_id / workflow_id / audit_trail_id	Lets the system jump directly from a document to the exact task, workflow instance, or audit history that produced it
comment_thread_id / keywords[] / tags[]	Powers in-context collaboration and improves search relevance beyond just the document title

3.9 Study / Country / Site Structural Fields

Field	What it does / why it matters
study_id, protocol_number, study_title, phase, therapeutic_area, sponsor, indication, study_status, start_date, end_date	Defines the top of the hierarchy — every zone, section, artifact, country, and site in the system ultimately belongs to one study record
country_id, study_id, country_code, regulatory_authority, ec_irb_name	Defines the country layer, which is what drives country-specific retention periods and which regulatory authority's rules apply
site_id, country_id, site_number, pi_name, site_status, activation_date, closure_date	Defines the site layer — site_status is what the Expected Document List reads to know which documents should now exist for that site

3.10 Investigator & Site Personnel Fields

Field	What it does / why it matters
pi_name / pi_medical_license_number	Principal Investigator identity and credential
pi_cv_version / pi_cv_date	Curriculum vitae version control
financial_disclosure_status	FDA 1572-related financial disclosure form (FDA 3455)
delegation_log_id	Delegation of Authority Log linkage — who is authorized to do what
gcp_training_certificate_id / gcp_training_expiry_date	GCP certification tracking with renewal alerts
form_1572_status	US IND Statement of Investigator (or country-equivalent form) status
site_signature_log_id	Log of all authorized signatures/initials at the site

3.11 Safety & Pharmacovigilance Fields

Field	What it does / why it matters
sae_id / safety_report_type	SAE / SUSAR / AE case identifier and classification
meddra_code	MedDRA-coded adverse event term
causality_assessment / expectedness	Investigator/sponsor causality judgment
reporting_timeline_status	24-hour / 7-day / 15-day expedited reporting compliance flag
dsmb_dmc_meeting_id / dsmb_report_date	Data Safety Monitoring Board oversight records

3.12 Investigational Product (IP) / Supply Chain Fields

Field	What it does / why it matters
ip_batch_lot_number / ip_expiry_date	Drug/device batch traceability
ip_storage_condition / temperature_excursion_flag	Cold-chain compliance
shipment_id / receipt_confirmation_date	Chain of custody
drug_accountability_log_id	Dispensing/return reconciliation
ip_return_destruction_record_id	End-of-study product disposition

3.13 Protocol & Amendment Fields

Field	What it does / why it matters
protocol_version_number	Current governing version
amendment_number / amendment_type	Substantial vs non-substantial, Global vs country-specific
amendment_reason / amendment_approval_date	Regulatory justification and sign-off

3.14 Informed Consent Fields

Field	What it does / why it matters
icf_version / icf_language	Consent form version and translation tracking
icf_type	Main / Assent (pediatric) / Legally Authorized Representative (LAR)
reconsent_required_flag / reconsent_date	Protocol-amendment-driven re-consent tracking
subject_pseudo_id	Pseudonymized reference only — never store direct subject identifiers in TMF

3.15 Regulatory Submission Fields

Field	What it does / why it matters
ind_cta_number	IND (US) / CTA (EU and others) reference number
submission_date / regulatory_authority_name	Health authority submission tracking
approval_reference_number	Authority approval/acknowledgement number
import_export_license_number	Cross-border IP shipment compliance

3.16 Monitoring Visit Fields

Field	What it does / why it matters
visit_type	Site Initiation (SIV) / Interim Monitoring (IMV) / Close-out (COV) / Remote

Field	What it does / why it matters
visit_date / monitor_name / visit_report_id	Visit record and linked report document
follow_up_letter_id	Post-visit correspondence
finding_id / capa_id / capa_status	Findings and Corrective/Preventive Action tracking

3.17 Vendor / CRO Oversight Fields

Field	What it does / why it matters
vendor_id / vendor_qualification_status	Third-party vendor qualification per ICH E6(R2) 5.2
oversight_plan_id / contract_id	Sponsor oversight documentation
transfer_of_obligations_flag	Which regulatory duties are delegated to CRO/vendor

3.18 Financial & Contractual Fields

Field	What it does / why it matters
site_budget_id / payment_schedule_id	Site payment tracking
clinical_trial_agreement_id	CTA contract reference
insurance_certificate_id / indemnity_document_id	Trial insurance and indemnification records

3.19 System / Multi-Tenant & Configuration Fields

Field	What it does / why it matters
tenant_id / organization_id	Multi-sponsor/multi-CRO tenant isolation
environment_type	Production / UAT / Sandbox / Training
custom_field_definitions[]	Sponsor-configurable metadata fields per study
study_template_id	Reusable Zone/Section/Artifact template per study type

3.20 AI & Automation Fields

Field	What it does / why it matters
classification_confidence_score	ML confidence for auto-suggested document type
suggested_zone_id / section_id / artifact_id	AI-recommended TMF placement, pending human confirmation
auto_tag_list[] / ocr_confidence_score	Automated indexing and text-extraction confidence

3.21 Sharing, Notification & Legal Hold Fields

Field	What it does / why it matters
external_share_link / share_expiry_date / watermark_flag	Controlled external collaboration
reminder_frequency_days / escalation_threshold_days	Configurable SLA and reminder rules
legal_hold_flag / legal_hold_reason / legal_hold_by / legal_hold_date	Prevents deletion during litigation/audit even past retention expiry

3.22 Training Record Fields

Field	What it does / why it matters
training_id / course_name / completion_date	Staff training history (GCP, protocol-specific, system training)
certificate_id / renewal_due_date	Certification proof and expiry alerts

3.23 Risk-Based Monitoring / Quality-by-Design Fields

Field	What it does / why it matters
risk_indicator_id / risk_level	Site/study risk signals (Low/Medium/High)
critical_to_quality_factor_flag	Marks data/documents critical to trial quality (ICH E8(R1) concept)

3.24 Attachment / Parent-Child Fields

Field	What it does / why it matters
parent_document_id / is_attachment_flag	Cover letter + attachment relationships
attachment_type	Appendix, exhibit, supporting evidence, etc.

3.25 Digital Signature Certificate Fields

Field	What it does / why it matters
certificate_id / signature_algorithm	Cryptographic identity behind an e-signature
timestamp_authority / signature_manifest_hash	Trusted timestamping and tamper-evidence binding

3.26 Version Control & Document Locking Fields

Field	What it does / why it matters
locked_by_user_id	Stores which user currently has the document checked out — blocks every other user from editing it at the same time and prevents two people from silently overwriting each other's work

Field	What it does / why it matters
lock_timestamp / lock_expiry	Records when the lock was taken and when it auto-releases, so a forgotten checkout does not block the document forever
checkout_status	Checked Out / Checked In / Read-Only — drives whether the Edit button is even clickable in the UI
version_diff_reference	Points to the previous version so the system can generate a side-by-side redline/diff view between any two versions
translation_group_id	Groups all language versions of the same source document together so a French ICF and its English source are recognized as one logical document
translation_language / is_translation_of_document_id	Identifies the language of this specific file and which original document it was translated from

3.27 Blinding & Randomization Fields (critical for GCP compliance)

Field	What it does / why it matters
blinding_status	Blinded / Unblinded / Open-label — controls whether a document is visible to blinded study team members at all, regardless of their normal role permissions
randomization_number	Subject's randomization code, stored only in the restricted unblinded module, never exposed to blinded roles
restricted_access_group	Blinded Team vs Unblinded Team — a second, independent permission layer that sits on top of RBAC so a Sponsor Admin who is blinded still cannot open unblinding documents
unblinding_reason / unblinding_date / unblinding_approver_id	Captures why, when, and who approved an emergency or planned unblinding event
emergency_unblinding_flag	Marks a document tied to an emergency code-break, which triggers mandatory notification to the medical monitor and an audit trail entry independent of normal document actions

4. Complete Feature List

4.1 Document & Content Management

- Drag-drop bulk upload plus guided single-upload wizard
- AI-assisted document type and TMF artifact classification
- Full version control with rollback
- Duplicate detection via hash comparison
- OCR and full-text search across all documents
- Bulk metadata edit
- Inline document preview (no download required)
- Placeholder document creation and tracking
- Expected Document List (EDL) engine — auto-generates what should exist per study/country/site/milestone

4.2 Quality Control & Compliance

- Configurable QC checklist per document type
- Multi-level QC workflow (Reviewer to QC Lead to Final sign-off)
- TMF completeness percentage by study/country/site/zone
- TMF Health Index (RAG scoring)
- Missing document, misfiled document, and duplicate reports
- ALCOA+ data integrity checks

4.3 Workflow & Task Management

- Configurable workflow engine (Draft to Review to Approve to Effective to Archive)
- Task assignment with email and in-app reminders
- Parallel and sequential approval routing
- SLA breach escalation rules
- Bulk workflow actions
- Delegate / out-of-office reassignment

4.4 Electronic Signature (21 CFR Part 11)

- In-app e-signature capture with meaning statement, identity, and timestamp
- Signature manifestation on the document
- Tamper-evident signature-to-document binding
- Multi-signer sequential or parallel signing
- Optional integration with DocuSign / Adobe Sign

4.5 Audit Trail

- Immutable log of every action: view, upload, edit, delete, download, status change, QC action, signature
- Captures who, what, when, old value, new value, IP address, and reason for change
- Read-only and non-editable
- Exportable for regulatory inspection

4.6 Access Control & Security

- Role-Based Access Control: Sponsor Admin, CRA, CRO User, Site Coordinator, Investigator, QC Reviewer, Auditor (read-only), System Admin
- Study, country, site, and zone-level permission scoping
- Multi-Factor Authentication
- SSO/SAML support
- IP whitelisting and session timeout controls
- Inspector read-only portal for regulatory auditors

4.7 Milestone & Study Lifecycle Tracking

- Study setup wizard with template-driven Zone/Section/Artifact population
- Milestone tracker: Protocol Approval, IRB/EC Approval, Site Activation, First Patient In, Last Patient Out, Database Lock, Study Closure
- Document-to-milestone linkage
- Country/site activation timeline view

4.8 Reporting & Analytics

- Completeness percentage by study/country/site/zone/section
- Filing timeliness and lag metrics
- QC pass/fail rates
- Outstanding placeholder report
- Inspection readiness scorecard
- Exportable reports in PDF, Excel, and CSV

4.9 Archival & Retention

- Auto-calculated retention period per country regulation
- Legal hold flag that overrides retention expiry
- Archive/lock study to read-only after closure
- Full TMF package export with index, documents, and audit trail

4.10 Collaboration

- Comment threads on documents
- @mention notifications
- Shared task inbox
- Limited-access site/CRO external portal

4.11 Integration Layer

- CTMS integration for site and monitoring data sync
- EDC integration
- eTMF to eISF (Investigator Site File) sync
- E-signature provider APIs
- SSO/identity provider integration
- REST API for all core objects

4.12 Mobile & Accessibility

- Responsive design for tablet-based site use
- Mobile upload from site visits
- WCAG 2.1 AA accessibility compliance

5. Vendor Feature Comparison

Feature	Veeva Vault	Medidata	ArisGlobal	Phlexglobal	Montrium
TMF Reference Model native	Yes	Yes	Yes	Yes	Yes
AI auto-classification	Yes (strong)	Partial	Yes (strong)	Partial	Limited
TMF Health/Quality Index	Yes	Yes	Yes	Yes (PhlexView)	Yes
Built-in QC workflow	Yes	Yes	Yes	Yes	Yes
E-signature (Part 11)	Yes	Yes	Yes	Yes	Yes (SharePoint-based)
CTMS integration	Yes (Vault CTMS)	Yes (Rave)	Yes	Yes	Via connectors
Inspector read-only mode	Yes	Yes	Yes	Yes	Yes
Platform base	Proprietary Vault	Cloud SaaS	Cloud SaaS	Cloud SaaS	Microsoft SharePoint
Known for	Market leader, deep config	EDC/CTMS ecosystem tie-in	AI document intelligence	Outsourced TMF services	SMB-friendly, familiar UI

Teaching insight: UI differs across vendors but the underlying data model — TMF Reference Model, audit trail, QC, and e-signature — is identical. Master this blueprint and you can operate any of these platforms.

6. Technical Architecture (React JS + Node JS)

6.1 Technology Stack

- **Frontend:** React JS (Vite), Redux Toolkit / React Query, Tailwind CSS, React Router, React Hook Form + Zod, Recharts, PDF.js
- **Backend:** Node JS + Express or NestJS, REST + optional GraphQL
- **Database:** PostgreSQL for relational/audit integrity + Redis for cache/session
- **File Storage:** AWS S3 or MinIO (S3-compatible)
- **Search:** Elasticsearch for full-text and metadata search
- **Auth:** JWT + refresh tokens, TOTP MFA, optional SAML SSO via Passport.js
- **E-signature:** Custom Part 11-style capture module, optional DocuSign API
- **Audit Logging:** Append-only Postgres table, optional Kafka event log for advanced teaching
- **File Processing:** Tesseract.js/AWS Textract for OCR, Puppeteer/PDFKit for PDF generation
- **DevOps:** Docker, GitHub Actions CI/CD, environment-based config

6.2 Core Database Tables

```
users, roles, permissions, studies, countries, sites, tmf_zones, tmf_sections, tmf_artifacts,
documents, document_versions, document_metadata, qc_checklists, qc_reviews, workflows,
workflow_steps, tasks, esignatures, audit_logs, milestones, study_milestones, comments,
notifications, retention_policies, expected_document_list, investigators, delegation_logs,
safety_reports, ip_shipments, monitoring_visits, vendors, training_records, legal_holds
```

6.3 Sample Core API Endpoints

```
POST /api/auth/login (+MFA step)
GET /api/studies
POST /api/studies
GET /api/studies/:id/zones
GET /api/studies/:id/tmf-completeness
POST /api/documents/upload
GET /api/documents/:id
PUT /api/documents/:id/metadata
POST /api/documents/:id/version
POST /api/documents/:id/qc-review
POST /api/documents/:id/sign
GET /api/documents/:id/audit-trail
GET /api/expected-document-list/:studyId
POST /api/workflows/:id/action
GET /api/reports/health-index/:studyId
GET /api/reports/export/:studyId
```

7. UI/UX Design System — Building a Top-Tier Interface

This is the design system your students should follow so the product feels like a real enterprise SaaS platform (Veeva Vault / Linear / Notion-grade), not a student project.

7.1 Design Philosophy

- Clean, data-dense enterprise SaaS aesthetic — information-rich without feeling cluttered
- Flat surfaces, no heavy gradients or drop shadows; rely on spacing and hierarchy, not decoration
- Consistency over novelty — every screen reuses the same component patterns
- Status communicated through color (RAG) plus icon, never color alone, for accessibility

7.2 Layout Structure

- **Left sidebar navigation:** collapsible, icon + label, sections for Dashboard, TMF Navigator, QC Queue, Tasks, Reports, Audit Trail, Admin
- **Top bar:** breadcrumb (Study > Country > Site), global search, notification bell, user/profile menu
- **Main content area:** context-aware — dashboard widgets, document grid, tree navigator, or detail panel depending on route
- **Right-side detail panel:** used in document viewer for metadata, version history, audit trail, and comments tabs without leaving the page

7.3 Color System

Field	What it does / why it matters
Primary/Accent	Deep blue — primary buttons, links, active nav state
Neutral grays	Backgrounds, borders, secondary text — majority of the UI surface
Green (RAG - Good)	Completeness above threshold, QC passed, on-time filing
Amber (RAG - Warning)	Approaching deadline, minor QC issues, pending review
Red (RAG - Critical)	Missing essential documents, overdue, QC failed
Purple	AI/automation-related indicators (auto-classification suggestions)

Rule: never rely on color alone for status — always pair with an icon and a text label for accessibility (WCAG 2.1 AA).

7.4 Typography

- Heading scale: H1 24-28px / H2 20px / H3 16-18px, medium weight only (no heavy bold)
- Body text: 14-16px, regular weight, 1.5-1.7 line height for readability in dense tables
- Monospace font for IDs, hashes, API paths, and technical metadata
- Sentence case everywhere — no ALL CAPS headers, no Title Case buttons

7.5 Core Component Library

- **Status badges/pills:** Rounded, colored background, dark text of same hue — used for document status, QC status, RAG health
- **Data tables:** Sortable columns, sticky header, row hover state, inline status badges, bulk-select checkboxes
- **Tree navigator:** Expandable Zone > Section > Artifact tree with document counts and health color per node
- **Metric/stat cards:** Muted label above, large number below — used on dashboards for completeness %, QC pass rate, filing lag

- **Progress/health bars:** Horizontal bar per zone/section showing RAG-colored completeness
- **Modals:** Upload wizard, e-signature capture, confirmation dialogs
- **Stepper:** Multi-step upload wizard and study setup wizard
- **Toast notifications:** Non-blocking confirmations (document uploaded, signature applied)
- **Empty states:** Friendly guidance with a clear action, not just a blank page
- **Skeleton loaders:** Used while documents/metadata load, never a blank spinner-only screen

7.6 Key Screen UI Patterns

- **Dashboard:** top row of 4 metric cards (Completeness, QC pass rate, Filing lag, Open queries), followed by a zone-by-zone health bar list, followed by a recent activity feed
- **Document viewer:** split layout — PDF/image preview on the left, tabbed metadata panel on the right (Details, Version History, QC, Audit Trail, Comments)
- **TMF navigator:** left tree panel (Zone > Section > Artifact) with a document grid/list on the right, filterable by status, country, and site
- **QC review queue:** list of pending items with inline checklist, pass/fail/query actions, and reviewer comments
- **Upload wizard:** step 1 select files, step 2 AI-suggested metadata (editable), step 3 confirm and submit for QC

7.7 Responsive Breakpoints

Field	What it does / why it matters
Desktop (>= 1280px)	Full sidebar + tree navigator + detail panel, three-column layouts
Tablet (768-1279px)	Collapsible sidebar, two-column layouts, detail panel becomes a slide-over
Mobile (< 768px)	Bottom navigation, single-column stacked layout, used mainly for site staff document uploads on the go

7.8 Accessibility & States

- WCAG 2.1 AA color contrast on all text and status indicators
- Full keyboard navigation and visible focus rings on every interactive element
- Screen-reader labels on icon-only buttons
- Explicit loading, empty, and error states designed for every screen — never a blank white page
- Dark mode supported via design tokens/CSS variables, not hardcoded colors

7.9 Design Tokens (for consistent implementation)

Field	What it does / why it matters
Spacing scale	4 / 8 / 12 / 16 / 24 / 32 / 48 px
Border radius	6px controls, 12px cards
Border	1px solid neutral-200 (light mode), neutral-700 (dark mode)
Shadow	Minimal — used only for modals/popovers, not flat cards
Icon set	Consistent outline icon library across the entire app (e.g. Tabler or Lucide icons)

8. Version Control — Detailed Mechanics

This is the part students most often build too shallow. A real eTMF version control system is not just 'save a new file' — it has locking, comparison, and a strict numbering rule.

8.1 Version Numbering Rule

- **Draft versions:** 0.1, 0.2, 0.3... — increment automatically on every save while status is Draft or In Review
- **Effective/Approved version:** jumps to 1.0 the moment the document is approved and signed — this is the version that counts in completeness
- **Minor revision after approval:** 1.1, 1.2 — small corrections that do not need full re-approval
- **Major amendment:** 2.0, 3.0 — a substantive change that requires the full review-approve-sign workflow again

8.2 Check-in / Check-out (Document Locking)

Prevents two people editing the same document at once, which is the single most common way real TMFs end up with conflicting versions.

- A user clicks 'Check out' — the system sets locked_by_user_id and checkout_status = Checked Out
- Every other user now sees the document as read-only with a banner showing who has it locked and since when
- On 'Check in', the system creates a new version, releases the lock, and writes an audit trail entry
- An auto-release timer (lock_expiry) frees a lock that was forgotten, so one absent-minded user cannot block a whole team indefinitely

8.3 Version Comparison (Diff View)

- Any two versions of the same document can be selected and compared side by side
- Text-based documents show a redline (added text green, removed text red); scanned/image documents show a side-by-side page viewer instead
- The diff view is itself logged in the audit trail as a 'version compared' action

8.4 Translation & Multi-Language Versioning

Translated documents (e.g. an ICF in French, German, and English) are linked through translation_group_id so they are managed as one family — approving the source document flags all its translations as needing re-review, instead of them silently going stale.

9. Access Control & Security — Detailed Mechanics

Access control in a real eTMF is not one flat role list — it is multiple layers stacked on top of each other. A user only sees a document if they pass every layer.

9.1 Layer 1 — Role-Based Access Control (RBAC) Matrix

Role	View	Upload	Edit	QC/Approve	Sign	Delete	Admin
Sponsor Admin	Yes	Yes	Yes	Yes	Yes	Yes	Yes
CRA / Monitor	Yes	Yes	Yes	No	No	No	No
CRO User	Yes	Yes	Yes	No	No	No	No
QC Reviewer	Yes	No	No	Yes	No	No	No
Site Coordinator	Site only	Yes	Yes	No	No	No	No
Investigator	Site only	Yes	No	No	Yes	No	No
Auditor/Inspector	Read-only	No	No	No	No	No	No
System Admin	Yes	No	No	No	No	No	Yes

9.2 Layer 2 — Hierarchical Scoping

Every permission from the RBAC matrix is further narrowed by study, country, site, and zone. A CRA assigned only to France cannot see Germany's documents even though their role technically allows 'Edit' — the scope layer filters what that role can act on.

9.3 Layer 3 — Blinding Controls (independent of RBAC)

This layer exists specifically because clinical trials are often randomized and blinded — a Sponsor Admin's normal role permissions must still not let them see unblinding documents if they are on the blinded team.

- Every user is tagged Blinded Team or Unblinded Team, separate from their functional role
- Documents flagged blinding_status = Unblinded are invisible to blinded-team users regardless of their RBAC role
- Emergency unblinding requires a documented reason, a named approver, and fires an immediate notification to the medical monitor

9.4 Layer 4 — Segregation of Duties

- The system blocks a user from QC-approving a document they themselves uploaded or authored
- The system blocks a single user from being both the preparer and the final approver on a workflow

9.5 Account & Session Security

- **Provisioning/deprovisioning:** access is granted through a request-and-manager-approval workflow, and automatically revoked the moment a user's study role ends or they leave the organization
- **Periodic access review:** a scheduled recertification (e.g. every 90 days) where study leads must actively confirm each user still needs their current access, or it is auto-revoked
- **Password & MFA policy:** minimum complexity, mandatory expiry, and MFA enforced for any role above read-only
- **Session controls:** auto-timeout after inactivity, concurrent-session limits, and optional IP/device restriction for high-sensitivity roles
- **Break-glass emergency access:** a rare, heavily audited path to grant temporary emergency access, which always requires a documented reason and triggers an immediate alert to the system owner

- **Print/export/download control:** downloads and prints can be disabled per confidentiality level, and allowed exports are dynamically watermarked with the downloading user's name, ID, and timestamp

10. Computer System Validation (CSV) & Regulatory Compliance Documentation

A GxP system is not compliant just because it has the right features — it must be proven, in writing, to work correctly. This is the part most student projects skip entirely, but it is exactly what a real sponsor's Quality/IT team will ask for.

- Validation Master Plan — defines the overall approach and scope for validating the eTMF system
- **IQ (Installation Qualification):** documented proof the system was installed correctly in its intended environment
- **OQ (Operational Qualification):** documented proof each feature works as specified — e.g. that the audit trail really does capture every field change
- **PQ (Performance Qualification):** documented proof the system performs correctly under real, intended day-to-day use
- 21 CFR Part 11 compliance checklist — a traceability matrix mapping each Part 11 requirement to the exact feature that satisfies it
- Change control process — any code or configuration change to a live system goes through impact assessment, testing, and approval before release
- Periodic review / revalidation — scheduled re-confirmation that the validated state still holds after updates
- User Acceptance Testing (UAT) sign-off, stored as a controlled document within the eTMF's own admin records

11. Backup, Disaster Recovery & Business Continuity

- Automated daily backups with defined retention, stored in a separate region/location from primary storage
- **RPO (Recovery Point Objective):** the maximum acceptable data loss window, e.g. 15 minutes — drives backup frequency
- **RTO (Recovery Time Objective):** the maximum acceptable downtime before the system must be restored, e.g. 4 hours
- Documented and periodically tested disaster recovery runbook — a backup that has never been restored in a drill is not a real backup
- Legacy TMF migration tooling — bulk import utility that preserves original metadata, dates, and audit history when a sponsor moves from paper or another system into this eTMF

12. Screen / Page List

- Login + MFA screen
- Dashboard — TMF health RAG heatmap, completeness %, pending tasks widget
- Study Setup Wizard — auto-populate Zones/Sections/Artifacts from template
- Country & Site Management
- Zone/Section/Artifact Navigator (tree view)
- Document Upload Wizard — drag-drop with AI metadata suggestion
- Document Detail/Viewer — preview, metadata, version history, audit trail tabs
- QC Review Queue & Checklist screen
- Workflow/Task Inbox
- E-Signature Modal
- Milestone Tracker (timeline/Gantt view)
- Expected Document List (EDL) manager
- Reports & Analytics dashboard
- Audit Trail Viewer
- User & Role Management (RBAC configuration)
- Archive/Export Study module
- Global Search (full-text + metadata filters)
- Inspector/Auditor Read-Only Portal
- Investigator & Delegation Log management
- Safety Reporting module (SAE/SUSAR tracking)
- IP/Supply Chain tracking module
- Vendor/CRO Oversight module
- Training Record management
- Version Diff/Compare Viewer
- Blinding & Unblinding Access Management (restricted admin screen)
- Access Review / Recertification screen
- System Validation (IQ/OQ/PQ) document repository

13. Phase-Wise Teaching Roadmap

- **Phase 1 — Foundations (Week 1-2):** TMF Reference Model, ICH GCP essential documents, ALCOA+ principles. Set up React+Node skeleton, DB schema, auth.
- **Phase 2 — Core Document Management (Week 3-4):** Study/country/site hierarchy, Zone/Section/Artifact navigator, upload, metadata, versioning.
- **Phase 3 — QC & Workflow (Week 5-6):** QC checklist engine, workflow states, task assignment, notifications.
- **Phase 4 — Compliance Layer (Week 7-8):** Audit trail middleware, e-signature module, RBAC, MFA — map every action to 21 CFR Part 11.
- **Phase 5 — Analytics & Reporting (Week 9):** TMF Health Index, completeness dashboard, filing-lag reports, exports.
- **Phase 6 — Advanced/Enterprise Features (Week 10-12):** Expected Document List engine, milestone tracking, mock CTMS/EDC integration, AI-assisted classification, archival/retention automation, inspector read-only mode, safety/IP/vendor modules.
- **Phase 7 — Version Control, Blinding & Validation (Week 13-14):** Check-in/check-out locking, version diff viewer, blinding/unblinding access layer, and a basic IQ/OQ/PQ validation documentation exercise.

Capstone Project

Students run a mock inspection: an 'FDA inspector' role logs in, searches for a specific essential document, checks its audit trail, verifies the e-signature, and confirms TMF completeness for a site — exactly like a real audit.

14. Key Teaching Emphasis Points

- Every field exists because of a real regulatory reason — tie fields back to GCP/Part 11/ALCOA+, not just because a vendor has it.
- Audit trail and e-signature are non-negotiable core, not optional add-ons.
- TMF completeness is measured against the Expected Document List, which is dynamic per study/country/site/milestone — not a raw document count.
- 'Certified copy' vs 'original' and 'placeholder' are TMF-specific concepts students must deeply understand.
- Inspection-readiness is the ultimate KPI of any eTMF — every feature should be built with the mindset: would this survive an FDA/EMA inspection?